

Method and provenance / 方法与来源记录

Status: READY_FOR_PREPROCESSING. The user directly authorized execution of this demo and supplied the pressure, composition, gate, capacity, expansion, and effective-heat basis.

状态: READY_FOR_PREPROCESSING。用户已直接授权运行该 demo，并给出了压力、组成、门控、吸附量、膨胀和有效热参数。

Sources searched:

- User-provided operating specification in the conversation.
- Local skills/idaes-process-modeler/references/psa.md and references/validation.md for cycle metrics and evidence boundaries.
- SciPy solve_ivp documentation: https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html for the BDF implicit variable-order integration method.

检索来源: 用户参数; 本插件关于 PSA 循环和验证边界的本地参考; SciPy solve_ivp 官方文档, 用于 BDF 隐式变阶积分器的数值说明。

Chosen calculation method: a component-inventory, finite-conductance, nonisothermal mixed-tank model. The raw machine-readable solver samples are saved in each case profiles.csv; derived comparison values are saved in comparison.csv. No smoothing, fitted measurement, or invented error bar is applied.

选择的计算方法: 组分库存 + 有限阀导纳 + 非等温全混罐模型。每个工况的原始求解采样保存在 profiles.csv, 派生对比值保存在 comparison.csv。不进行平滑、不把模拟当拟合实验、也不添加虚构误差棒。

Assumptions: 298 K reference state; STP = 273.15 K and 101325 Pa; isotropic 5% solid expansion; 10 μm reference crystal length; 15 kJ mol^{-1} diffusion activation energy; 0.02/0.05 s^{-1} CO₂/N₂ LDF coefficients; 0.1 s^{-1} structural relaxation; assumed heat capacities and UA; 40 kJ mol^{-1} as one effective CO₂ adsorption/structural heat magnitude; no measured hysteresis loop; no competitive mixture fit; no vacuum-pump power calculation.

假设包括: 298 K 参考状态; STP 为 273.15 K、101325 Pa; 各向同性 5% 实体膨胀; 10 μm 参考晶体长度; 15 kJ mol^{-1} 扩散活化能; CO₂/N₂ 的 LDF 系数为 0.02/0.05 s^{-1} ; 结构响应速率为 0.1 s^{-1} ; 热容和 UA 为假设值; 40 kJ mol^{-1} 作为一次性的 CO₂ 吸附/结构有效热量; 没有实测滞后环、竞争吸附拟合和真空泵功耗计算。

Rejected alternatives: fixed 15% CO₂ composition during evacuation; purge-gas regeneration; counting gas initially present in the vessel as recovered solid adsorbate; adding the 40 kJ mol^{-1} value twice; calling CSS or balance closure physical validation.

明确排除: 抽气期间固定 15% CO₂ 组成; 混合气吹扫再生; 把容器初始气体当作固相捕集量; 重复加入 40 kJ mol^{-1} ; 把 CSS 或方程闭合称为物理验证。